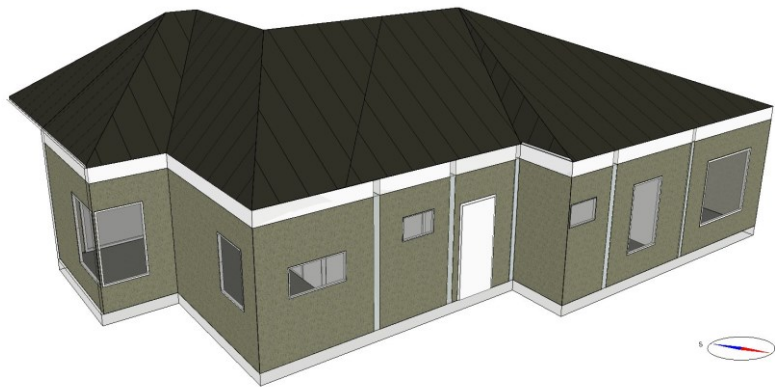


Low-Energy Family House

Thermal Modelling Package

IDA ICE Model, State-Space Model and Weather Data — Student Handout



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How to cite this model

Any work using the IDA ICE model, the state-space model or the accompanying weather data must cite the following two references:

[1] A. Banjac and D. Prvonožec, “Optimizing Insulation Thickness for Energy and Cost Efficiency in Residential Buildings: A Case Study,” Proc. 11th Int. Conf. on Control, Decision and Information Technologies (CoDIT), 2025, pp. 1630–1635. doi: 10.1109/CoDIT66093.2025.11321719

[2] D. Prvonožec and A. Banjac, “Thermal Comfort–Based vs Temperature–Based Control in a Single-Family House: Impacts on Energy Use and Occupant Comfort,” SpliTech 2026.

Reference [1] contains a detailed description of the building (geometry, envelope, materials, energy performance). **Reference [2]** describes the mathematical (state-space) model and a sample PMV-based control application.

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1. Overview and Purpose

This document accompanies a thermal modelling package for a representative low-energy single-family house under the continental climate of the Zagreb region, Croatia. The package contains two models of the same building at two different levels of detail, plus the matching weather inputs:

- a high-fidelity IDA ICE building model (Equa Simulation AB) that resolves the full 3-D geometry, materials and HVAC system — bundled together with its TMY weather file in a single self-contained zip;
- a reduced-order linear state-space (SS) model identified from IDA ICE simulation data, suitable for fast simulation and for control design (PID, MPC);
- the matching weather year for the Zagreb region, provided as a MATLAB timeseries (ExtCond_Zagreb.mat) for use with the state-space model. The same weather year is included as a .prn climate file inside the IDA ICE zip.

The package is intended as a foundation for course assignments, student projects and research on building thermal dynamics, control strategies, energy management and thermal comfort. The building characteristics, the energy performance, and the identified model are documented in [1] and [2]. The identification methodology follows [3].

Repository layout

```
low_energy_family_house/
├── README.md
├── docs/
│   └── low_energy_family_house_description.pdf    ← this document
├── model_ida_ice/
│   └── low_energy_family_house.zip                ← IDA ICE project + .prn TMY
├── model_ss/
│   ├── low_energy_family_house_ss_model.mat      ← SS matrices A, B, C, D
│   └── TMY_data_Zagreb.mat                       ← matching TMY for Simulink / MATLAB
```

The IDA ICE project zip in `model_ida_ice/` is **self-contained**: after unzipping, the `.idm` project file already points to its TMY weather file (`TMY_data_Zagreb_region.prn`) — no extra download is needed to run a whole-year simulation.

2. Building type and general characteristics

The modelled building is a representative low-energy, detached, single-storey family house designed for the continental climate of the Zagreb region (approx. 45.7° N, 16.05° E, elevation ≈ 111 m). The model corresponds to a real building of this type whose energy performance has been documented in [1].

Key characteristics:

- Building type: detached single-storey family house (no upper floor, no basement)
- Construction: prefabricated, timber-frame technology
- Total internal floor area: 138.13 m²
- Internal clear ceiling height: 2.75 m
- Number of internal spaces: 12 (open-plan kitchen + dining + living room, four bedrooms, bathroom, WC, corridor, entrance vestibule, pantry, utility room)
- Energy classification: low-energy house (high envelope insulation)
- HVAC: air-to-water heat pump + underfloor heating; modelled in IDA ICE with an idealized four-pipe system allowing both heating and cooling year-round
- Climate: continental Croatian (cold winters, warm summers — see Section 6)

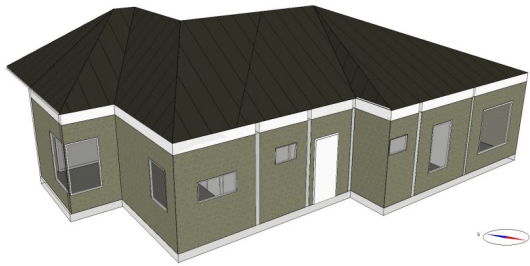


Figure 1 3-D view of the building model (IDA ICE rendering, view from south-west).

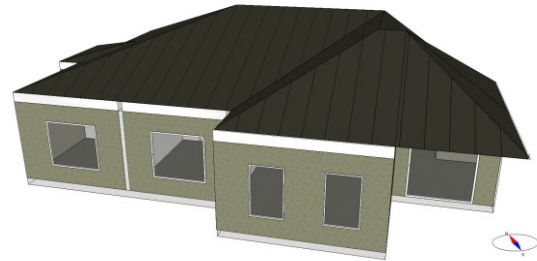


Figure 2 3-D view of the building model (IDA ICE rendering, view from north-east).

The external envelope meets passive-house-grade thermal performance [1]:

- External wall (total thickness ≈ 36 cm): timber-frame core + 30 cm mineral wool in the wall cavity + ETICS façade with 15 cm stone wool combined with 16 cm mineral wool
- Foundation slab: 16 cm extruded polystyrene (XPS) under the floor slab
- Windows and external doors: high-quality 6-chamber PVC profiles, triple-pane low-emissivity glazing filled with argon, three sealing levels
- Air-tightness and thermal bridging optimized for low-energy performance

The full layer-by-layer description of every construction element (external walls, internal walls, floor, ceiling/roof, windows) with material properties (thermal conductivity, density, specific heat, layer thickness) is encoded directly in the IDA ICE project file (see Section 3) and tabulated in [1].

The house has 12 internal spaces, with the floor areas given in Table 1. The orientation is shown by the north arrow in Figure 3: north is to the lower-left of the plan, so the main glazed wall of the kitchen + living room (the largest room, 52.05 m²) faces approximately north-west, and the long external wall of the bedrooms faces approximately east. The four external wall groups used in the identified state-space model are labelled E, N, S, W and refer to the orientation classes closest to each cardinal direction (the building is not exactly aligned with the cardinal axes; see Section 6.3).



Figure 3 Floor plan of the modelled house. Areas of the rooms are given in m^2 . The N arrow shows true north.

Table 1 lists every space, its area, and whether it is treated as a thermal zone with active heating/cooling in the reduced-order state-space model. Six conditioned spaces are aggregated into six thermal zones; the unconditioned auxiliary spaces (entrance vestibule, corridor, WC, pantry, utility room) are not separately modelled in the SS model — they are absorbed into the thermal mass of the adjacent conditioned zones during identification.

#	Room (Croatian)	Room (English)	Area [m^2]	SS zone
1	Blagovaonica / Dnevni boravak	Kitchen + Dining + Living	52.05	Zone 1
2	Spavaća soba 1 (B1)	Bedroom B1	15.55	Zone 2
3	Spavaća soba 2 (B2)	Bedroom B2	15.33	Zone 3
4	Spavaća soba 3 (B3)	Bedroom B3	10.12	Zone 4
5	Spavaća soba 4 (B4)	Bedroom B4	11.42	Zone 5
6	Kupaonica	Bathroom	8.13	Zone 6
7	Hodnik	Corridor	8.22	(part of Zone 1)
8	Vjetrobran / ulaz	Entrance vestibule	7.00	(part of Zone 1)
9	Tehnička prostorija	Utility room	4.97	(part of Zone 1)
10	WC	WC (toilet)	3.06	(part of Zone 1)
11	Izba	Pantry	2.28	(part of Zone 1)
12	Kuhinja (dio otvorenog tlocrta)	Kitchen (open-plan, part of Zone 1)	—	(part of Zone 1)

Note: in [1] the kitchen is sometimes listed as a separate space of $9.5 m^2$; in the SS model it is treated as part of the open-plan Kitchen/Dining/Living zone (total $\approx 52 m^2$).

3. IDA ICE high-fidelity model

IDA ICE (Indoor Climate and Energy, Equa Simulation AB) is a professional dynamic multi-zone building energy simulation tool. It resolves the full 3-D geometry of the building, layer-by-layer construction of every surface, glazing properties, HVAC components, occupancy and internal gains, and uses a detailed solver for the coupled heat-, air- and moisture-transfer equations. The IDA ICE model is the high-fidelity reference from which the simplified state-space model is identified.

Files in model_ida_ice/

low_energy_family_house.zip — self-contained IDA ICE project. Contains the .idm project file, the TMY weather file (TMY_data_Zagreb_region.prn) and all other referenced files (materials, components, schedules, geometry). Everything required to open and run the model in IDA ICE is inside this single zip.

4. Low order semi-physical building state-space model

Each conditioned room is modelled by two states:

- $T_{a,i}$ — air temperature in room i (fast dynamics, low thermal capacity)
- $T_{z,i}$ — thermal-mass temperature of room i (slow dynamics; aggregates walls, ceiling, floor and furniture)

The two-state per-room formulation used here follows equation (1) in [2]. In compact parametric form:

$$\begin{bmatrix} \dot{T}_{a,i} \\ \dot{T}_{z,i} \end{bmatrix} = \begin{bmatrix} -(p_{1,i} + p_{3,i} + p_{7,i}) & p_{1,i} \\ p_{2,i} & -p_{2,i} \end{bmatrix} \begin{bmatrix} T_{a,i} \\ T_{z,i} \end{bmatrix} + \begin{bmatrix} p_{3,i} & p_{4,i} & p_{5,i} & p_{6,i} \\ 0 & 0 & 0 & 0 \end{bmatrix} u$$

with per-room input vector

$$u = [T_{\text{out}} \quad I_{b,i} \quad I_{d,i} \quad P_{\text{hc},i}]^T,$$

Where T_{out} is the outdoor air temperature, $I_{b,i}$ and $I_{d,i}$ are the direct and diffuse solar irradiance incident on the room's external wall, and $P_{\text{hc},i}$ is the heating/cooling thermal power. Parameters $p_1 \dots p_6$ are identified from IDA ICE simulation data following [3]; they are combinations of the thermal resistances, heat capacities and solar gain coefficients of the room. The full theoretical formulation in [2] includes an additional term that represents heat exchange with adjacent rooms. In the implemented house model this term is set to zero because all conditioned rooms operate at similar setpoints (22 °C in winter, 24.5 °C in summer); the temperature difference between adjacent zones is therefore small and the inter-zone heat flux is a negligible contribution to the overall energy balance. Combining the six per-room models gives the full house state-space model in the standard linear time-invariant form:

$$\begin{aligned} \dot{x}(t) &= A \cdot x(t) + B \cdot u(t) \\ y(t) &= C \cdot x(t) + D \cdot u(t) \end{aligned}$$

Dimensions:

Matrix	Size	Property
A	12 × 12	block-diagonal structure (6 zone pairs)
B	12 × 15	external + control inputs
C	12 × 12	identity (all states are outputs)
D	12 × 15	zero (no direct feedthrough)

The state vector is structured as $x = [T_{a,1}, T_{a,2}, \dots, T_{a,6}, T_{z,1}, T_{z,2}, \dots, T_{z,6}]^T$: first the six air temperatures (fast states), then the six thermal-mass temperatures (slow states). All in °C. The mapping between zone number and room is fixed by the identification setup and matches the order used in the IDA ICE project.

The input vector $u \in \mathbb{R}^{15}$ groups the inputs into three categories: outdoor temperature, solar irradiance on the four external wall groups (direct and diffuse), and per-zone heating/cooling power. The first 9 inputs are the external (uncontrolled) inputs supplied by ExtCond_Zagreb.mat; the last 6 are the control variables, one per zone.

Index	Variable	Description	Unit
u_1	T_{out}	Outdoor air temperature	°C
u_2	I_b^E	Direct solar irradiance on East wall	W/m ²
u_3	I_b^N	Direct solar irradiance on North wall	W/m ²
u_4	I_b^S	Direct solar irradiance on South wall	W/m ²
u_5	I_b^W	Direct solar irradiance on West wall	W/m ²
u_6	I_d^E	Diffuse solar irradiance on East wall	W/m ²
u_7	I_d^N	Diffuse solar irradiance on North wall	W/m ²
u_8	I_d^S	Diffuse solar irradiance on South wall	W/m ²
u_9	I_d^W	Diffuse solar irradiance on West wall	W/m ²
u_{10}	$P_{\text{hc},1}$	Heating (+) / cooling (–) — Kitchen/Living	W
u_{11}	$P_{\text{hc},2}$	Heating / cooling — Bedroom B1	W
u_{12}	$P_{\text{hc},3}$	Heating / cooling — Bedroom B2	W
u_{13}	$P_{\text{hc},4}$	Heating / cooling — Bedroom B3	W
u_{14}	$P_{\text{hc},5}$	Heating / cooling — Bedroom B4	W
u_{15}	$P_{\text{hc},6}$	Heating / cooling — Bathroom	W

Recommended power limits in simulations: $-5000 \text{ W} \leq P_{\text{hc},i} \leq +5000 \text{ W}$ per zone (negative = cooling, positive = heating), as used in [2].

The output vector y equals the full state vector x . All 12 zone temperatures (6 air + 6 thermal-mass) are therefore directly available as outputs. In real installations $T_{z,i}$ is not directly measured and must be estimated (e.g., constrained Kalman filter [3]).

5. External Conditions (Weather Data)

Both the IDA ICE model and the state-space model are driven by the same weather year — a Typical Meteorological Year (TMY) for the Zagreb region, based on weather observations from 2005 to 2023 [7]. TMY data consist of hourly meteorological values for each hour of the year, specific to a given geographical location. The data were selected from hourly measurements taken over a longer period, with monthly data chosen from the year considered most "typical" for each month. For example, January might be from 2007, February from 2012, and so on. The climate data file contains measurements of air temperature, relative humidity, wind speed, diffuse horizontal radiation, and direct normal solar radiation.

The data is provided in two equivalent formats, one for each tool:

- **TMY_data_Zagreb_region.prn** - bundled inside the IDA ICE project zip
- **TMY_data_Zagreb.mat** — MATLAB timeseries object stored separately in model_ss/

Both files describe the same weather year; they differ only in format. The .prn file contains all variables IDA ICE needs (outdoor temperature, humidity, direct and diffuse horizontal solar radiation, wind speed and

direction, etc.). The .mat file contains only the 9 signals that act as inputs $u_1 \dots u_9$ of the state-space model, already projected onto the four external-wall orientations of the building.

References

- [1.] A. Banjac and D. Prvonožec, “Optimizing Insulation Thickness for Energy and Cost Efficiency in Residential Buildings: A Case Study,” Proc. 11th Int. Conf. on Control, Decision and Information Technologies (CoDIT), 2025, pp. 1630–1635. doi: 10.1109/CoDIT66093.2025.11321719
- [2.] D. Prvonožec and A. Banjac, “Thermal Comfort–Based vs Temperature–Based Control in a Single-Family House: Impacts on Energy Use and Occupant Comfort,” SplitTech 2026.
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- [4.] A. Banjac, H. Novak and M. Vašak, “Implementation of model predictive indoor climate control for hierarchical building energy management,” Control Engineering Practice, vol. 136, p. 105536, 2023.
- [5.] Equa Simulation AB, IDA-ICE Version 4.8, 2023.
- [6.] ISO 7730:2005, Ergonomics of the thermal environment — Analytical determination and interpretation of thermal comfort using the PMV and PPD indices, 2005.
- [7.] Joint Research Centre, “Photovoltaic geographical information system (PVGIS) - Typical Meteorological Year generator,” 2023, accessed: 2023-10-14. [Online]. Available: https://re.jrc.ec.europa.eu/pvg_tools/en/#TMY